

NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 2 EXAMINATION 2013-2014

EE7403 – IMAGE ANALYSIS AND PATTERN RECOGNITION

April / May 2014

Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 5 questions and comprises 5 pages.
2. Answer ALL questions.
3. All questions carry equal marks.
4. This is a closed-book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. A digital image $f(x, y)$ of size 7×7 is given in Figure 1.

0	3	0	0	0	0	0
3	9	3	3	3	9	3
0	3	0	0	0	0	0
0	3	0	0	0	0	0
0	9	0	0	3	9	3
0	3	0	0	3	9	3
0	3	0	0	3	3	3

Figure 1

- (a) A Linear Shift Invariant (LSI) filter of size 1×3 with mask $[1/3 \ 1/3 \ 1/3]$ is applied to image $f(x, y)$. Find the output image $g_1(x, y)$ of size 7×7 with the assumption that $f(x, y) = 0$ for pixels outside of Figure 1.
(7 Marks)
- (b) A median filter of size 1×3 is applied to image $f(x, y)$. Find the output image $g_2(x, y)$ of size 7×7 with the assumption that $f(x, y) = 0$ for pixels outside of Figure 1.
(7 Marks)

Note: Question No. 1 continues on page 2.

- (c) Suppose the image $f(x, y)$, which contains information of a horizontal line, a vertical line and a square, has been corrupted by noise. Discuss and compare the two filtered images $g_1(x, y)$ and $g_2(x, y)$. (6 Marks)
2. A binary image, whose black pixels constitute set A , is shown in Figure 2.1. A structure element (a filled circle), whose black pixels constitute set B , is shown in Figure 2.2. Assume that the origin of B is at its center. Note that the diameter of structure element B is larger than the diameter of the smallest circle and the width of the strip in A , and is smaller than the diameter of the two larger circles in A . The boundary of A is shown in Figure 2.3 on page 3.

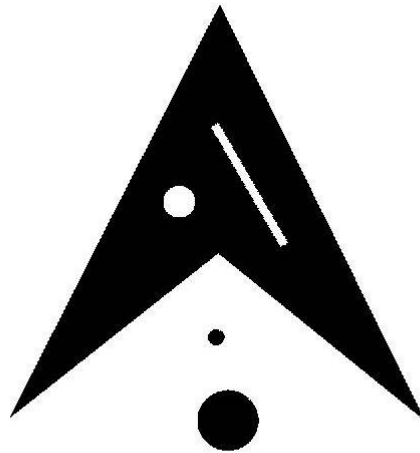


Figure 2.1



Figure 2.2

Note: Question No. 2 continues on page 3.

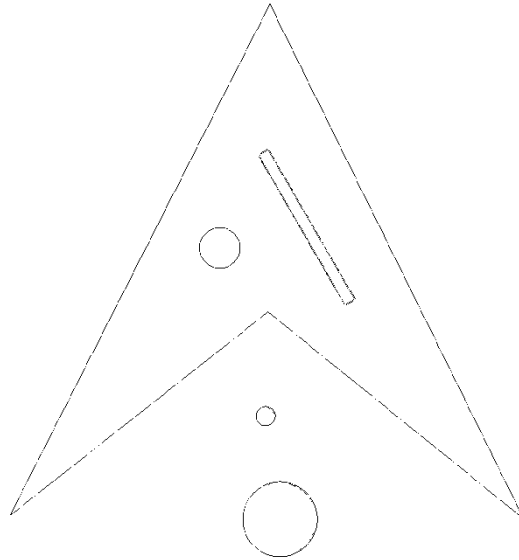


Figure 2.3

- (a) Sketch the boundary of A using dotted line (or pencil) and sketch the boundary of the dilation of A by B , $A \oplus B$ using solid line (or pen) in the same plot.
(5 Marks)
- (b) Sketch the boundary of A using dotted line (or pencil) and sketch the boundary of the erosion of A by B , $A \ominus B$ using solid line (or pen) in the same plot.
(5 Marks)
- (c) Sketch the boundary of A using dotted line (or pencil) and sketch the boundary of the closing of A by B , $A \bullet B$ using solid line (or pen) in the same plot.
(5 Marks)
- (d) Sketch the boundary of A using dotted line (or pencil) and sketch the boundary of the opening of A by B , $A \circ B$ using solid line (or pen) in the same plot.
(5 Marks)

3. A local image pattern is called an oriented pattern if its gray level is a constant along an orientation. A particular image is modeled locally by $f(x, y) = h(\omega x - \sqrt{3}\omega y)$, where $h(t)$ is a differentiable function, ω is a constant, and x and y are continuous variables.

- (a) Determine the orientation ϕ , where $-\frac{\pi}{2} < \phi \leq \frac{\pi}{2}$, along which, $f(x, y)$ is a constant.

(5 Marks)

- (b) Compute the orientation ϕ based on the direction of image gradient $\nabla f(x, y) = [\frac{\partial f(x, y)}{\partial x}, \frac{\partial f(x, y)}{\partial y}]$.

(5 Marks)

- (c) A discrete image $F(n, m)$ is obtained by sampling the image $f(x, y)$ with a sampling interval Δ . Given $h(t) = \sin(t)$, the discrete image is expressed as $F(n, m) = \sin(\omega\Delta m - \sqrt{3}\omega\Delta n)$. The gradient of the discrete image is approximated by the finite differences as

$$\nabla F(m, n) = [F(m+1, n) - F(m-1, n), F(m, n+1) - F(m, n-1)].$$

Estimate the image orientation using the approximated gradient $\nabla F(m, n)$ for $\Delta = \pi / (2\omega)$ and for $\Delta = \pi / (8\omega)$, respectively. Suggest how the estimation error can be reduced.

(10 Marks)

4. Let $p(\omega_i)$ be the prior probability of class ω_i and $p(\omega_i | \mathbf{x})$ be the posterior probability of class ω_i . Let $p(\mathbf{x} | \omega_i)$ be the class-conditional probability density function and $p(\mathbf{x})$ be the probability density function over all classes.

- (a) If you know the value of data $\mathbf{x}$ and make a decision that $\mathbf{x}$ belongs to a class ω_i , find the probability of making the wrong decision $p(e_i | \mathbf{x})$, and hence determine the decision rule that minimizes $p(e_i | \mathbf{x})$, using one of the above quantities $[p(\omega_i), p(\omega_i | \mathbf{x}), p(\mathbf{x} | \omega_i), p(\mathbf{x})]$.

(5 Marks)

- (b) Suppose none of the above quantities is available but we can use the available training data to estimate $p(\omega_i)$, $p(\mathbf{x} | \omega_i)$ and $p(\mathbf{x})$, derive the decision rule that minimizes the probability of making the wrong decision so that you can make a decision based on the estimated $p(\omega_i)$ and $p(\mathbf{x} | \omega_i)$.

(5 Marks)

Note: Question No. 4 continues on page 5.

- (c) Estimate $p(\omega_i)$, $p(\mathbf{x}|\omega_i)$ and $p(\mathbf{x})$ from the training data so that the decision rule that minimizes the probability of making the wrong decision leads to a simple k -nearest neighbor classifier.
(10 Marks)
5. Given a $m \times n$ data matrix $\mathbf{X}$ that contains n centralized training samples, i.e. the sample mean vector is zero.
- (a) Compute the covariance matrix Σ of the training data $\mathbf{X}$, and write down the definition of the eigenvalues λ_i and eigenvectors ϕ_i of the covariance matrix Σ . Then show that the variance σ_i^2 of the projected training data on the unit-length eigenvector ϕ_i is equal to the eigenvalue λ_i .
(8 Marks)
- (b) Prove that two eigenvectors ϕ_k , ϕ_j corresponding to two different eigenvalues λ_k , λ_j , where $\lambda_k \neq \lambda_j$, are orthogonal.
(6 Marks)
- (c) Find the covariance σ_{kj}^2 between the projected training data on the eigenvector ϕ_k and the projected training data on the eigenvector ϕ_j , where $\lambda_k \neq \lambda_j$.
(6 Marks)

END OF PAPER

3. A local image pattern is called an oriented pattern if its gray level is a constant along an orientation. A particular image is modeled locally by $f(x, y) = h(\omega x - \sqrt{3}\omega y)$, where $h(t)$ is a differentiable function, ω is a constant, and x and y are continuous variables.

- (a) Determine the orientation ϕ , where $-\frac{\pi}{2} < \phi \leq \frac{\pi}{2}$, along which, $f(x, y)$ is a constant.

(5 Marks)

$$(a) \quad f(x, y) = h(ax + by) = h(\omega x - \sqrt{3}\omega y)$$

$$\begin{aligned} \phi &= \tan^{-1} \frac{a}{b} = \tan^{-1} \left(\frac{\omega}{-\sqrt{3}\omega} \right) \\ &= \tan^{-1} \frac{-1}{\sqrt{3}} = -\frac{\pi}{6} \end{aligned}$$

- (b) Compute the orientation ϕ based on the direction of image gradient

$$\nabla f(x, y) = \left[\frac{\partial f(x, y)}{\partial x}, \frac{\partial f(x, y)}{\partial y} \right].$$

$$\frac{\partial f(x, y)}{\partial x} = \frac{\partial h}{\partial t} \omega$$

$$ax + by = t$$

(5 Marks)

$$\omega x - \sqrt{3}\omega y = t$$

$$\frac{\partial f(x, y)}{\partial y} = -\sqrt{3} \frac{\partial h}{\partial t} \omega$$

$$= \begin{bmatrix} \frac{\partial h}{\partial t} \omega \\ -\sqrt{3} \frac{\partial h}{\partial t} \omega \end{bmatrix}$$

$$\Rightarrow \phi = \tan^{-1} \frac{\frac{\partial f}{\partial y}}{\frac{\partial f}{\partial x}} = \tan^{-1} -\sqrt{3} = -\frac{\pi}{3}$$

- (c) A discrete image $F(n, m)$ is obtained by sampling the image $f(x, y)$ with a sampling interval Δ . Given $h(t) = \sin(t)$, the discrete image is expressed as $F(n, m) = \sin(\omega\Delta m - \sqrt{3}\omega\Delta n)$. The gradient of the discrete image is approximated by the finite differences as

$$\nabla F(m, n) = [F(m+1, n) - F(m-1, n), F(m, n+1) - F(m, n-1)].$$

Estimate the image orientation using the approximated gradient $\nabla F(m, n)$ for $\Delta = \pi / (2\omega)$ and for $\Delta = \pi / (8\omega)$, respectively. Suggest how the estimation error can be reduced.

(10 Marks)

$$\textcircled{1} \quad \Delta = \frac{\pi}{2\omega} \quad F(m, n) = \sin\left(\frac{\pi}{2}m - \frac{\sqrt{3}}{2}n\right)$$

$$\begin{aligned} \nabla F(m, n) &= \sin\left(\frac{\pi}{2}m + \frac{\pi}{2} - \frac{\sqrt{3}}{2}n\right) - \sin\left(\frac{\pi}{2}m - \frac{\pi}{2} - \frac{\sqrt{3}}{2}n\right) \\ &\quad + j \left[\sin\left(\frac{\pi}{2}m - \frac{\sqrt{3}}{2}n - \frac{\sqrt{3}}{2}\pi\right) - \sin\left(\frac{\pi}{2}m - \frac{\sqrt{3}}{2}n + \frac{\sqrt{3}}{2}\pi\right) \right] \end{aligned}$$

$$\textcircled{2} \quad \Delta = \frac{\pi}{8\omega} \quad F(m, n) = \sin\left(\frac{\pi}{8}m - \frac{\sqrt{3}}{8}\pi n\right)$$

$$\begin{aligned} \nabla F(m, n) &= \sin\left(\frac{\pi}{8}m + \frac{\pi}{8} - \frac{\sqrt{3}}{8}\pi n\right) - \sin\left(\frac{\pi}{8}m - \frac{\pi}{8} - \frac{\sqrt{3}}{8}\pi n\right) \\ &\quad + j \left[\sin\left(\frac{\pi}{8}m - \frac{\sqrt{3}}{8}\pi n - \frac{\sqrt{3}}{8}\pi\right) - \sin\left(\frac{\pi}{8}m - \frac{\sqrt{3}}{8}\pi n + \frac{\sqrt{3}}{8}\pi\right) \right] \end{aligned}$$

Suggest: we can use Gaussian - shaped smoothing to reduce the noise.
or increase sampling interval.

5. Given a $m \times n$ data matrix $\mathbf{X}$ that contains n centralized training samples, i.e. the sample mean vector is zero.

(a) Compute the covariance matrix Σ of the training data $\mathbf{X}$, and write down the definition of the eigenvalues λ_i and eigenvectors ϕ_i of the covariance matrix Σ . Then show that the variance σ_i^2 of the projected training data on the unit-length eigenvector ϕ_i is equal to the eigenvalue λ_i .

(8 Marks)

(b) Prove that two eigenvectors ϕ_k, ϕ_j corresponding to two different eigenvalues λ_k, λ_j , where $\lambda_k \neq \lambda_j$, are orthogonal.

(6 Marks)

(c) Find the covariance σ_{kj}^2 between the projected training data on the eigenvector ϕ_k and the projected training data on the eigenvector ϕ_j , where $\lambda_k \neq \lambda_j$.

(6 Marks)

(a) m features n samples $\mu = 0$

$$\Sigma = \mathbf{X} \mathbf{X}^T$$

$$\Sigma \phi_i = \lambda_i \phi_i$$

$$\phi_i^T \phi_i = \|\phi_i\|^2 = 1$$

$$\sigma_i^2 = \phi_i^T \Sigma \phi_i = \lambda_i \underbrace{\phi_i^T \phi_i}_1 = \lambda_i$$

$$(b) \quad \Sigma \phi_k = \lambda_k \phi_k \quad \Sigma \phi_j = \lambda_j \phi_j$$

$$\begin{aligned} \phi_j^T \Sigma \phi_k &= \phi_j^T \lambda_k \phi_k & (\Sigma \phi_j)^T \phi_k &= \phi_j^T \Sigma^T \phi_k \\ &= \lambda_k \phi_j^T \phi_k & &= (\lambda_j \phi_j)^T \phi_k \\ & & &= \lambda_j \phi_j^T \phi_k \end{aligned}$$

Due to Σ is symmetric matrix

$$\Sigma^T = \Sigma$$

$$\therefore \lambda_k \phi_j^T \phi_k = \lambda_j \phi_j^T \phi_k$$

$$(\lambda_k - \lambda_j) \phi_j^T \phi_k = 0 \quad \lambda_k \neq \lambda_j \Rightarrow \phi_j^T \phi_k = 0$$

- (c) Find the covariance σ_{kj}^2 between the projected training data on the eigenvector ϕ_k and the projected training data on the eigenvector ϕ_j , where $\lambda_k \neq \lambda_j$.

$$\begin{aligned}\sigma_k^2 &= \phi_k^T \Sigma \phi_k \\ &= \lambda_k\end{aligned}$$

$$\begin{aligned}\sigma_j^2 &= \phi_j^T \Sigma \phi_j \\ &= \lambda_j\end{aligned}$$

$$\lambda_k = \sum_{i=1}^n (x_i - \mu_k)^2$$

$$\lambda_j = \sum_{i=1}^n (x_i - \mu_j)^2$$

$$\sigma_{kj}^2 = (\mu_k - \mu_j)^2$$

Due to mean vector is 0

$$\mu_k = \mu_j = 0$$

$$\sigma_{kj}^2 = 0$$